

Question 1

$$V_m = 220 \text{ kV}, C_t = 2500 \text{ pF}, k = 2$$

$$V_t = 1.9 \cdot V_m = 418 \text{ kV} \rightarrow \text{a } 500 \text{ kV test transformer can be used}$$

$$P_t = k \omega C_t V_t^2 = 2 \cdot 2\pi 50 \text{ Hz} \cdot (2500 \text{ pF})^2 \cdot (418 \text{ kV})^2 \approx 274 \text{ kVA}$$

$\rightarrow 200 \text{ kVA}$ is not enough!

1.1) That 500 kV/200 kVA test transformer could withstand the voltage but not the power, hence **it CANNOT BE USED**

$$V_{out} = V_t = 418 \text{ kV} = n \cdot V_m \Rightarrow n = \frac{418 \text{ kV}}{220 \text{ kV}} = \frac{418}{220} = 1.9 \sim 2$$

$$\eta = \frac{2}{n+1} = \frac{2}{2+1} = 67\%$$

1.2) **Two transformers** is the minimum amount and the efficiency would be **67%**

$$f_0 = \frac{1}{2\pi \sqrt{L_t C_t}} \Rightarrow L_t = \frac{1}{C_t (2\pi f_0)^2} = \frac{1}{2500 \text{ pF} \cdot (2\pi \cdot 50 \text{ Hz})^2} = 4.053 \text{ mH}$$

$$Q = \frac{\sqrt{4C}}{R} = \frac{\sqrt{4.053 \text{ mH} / 2500 \text{ pF}}}{160 \text{ k}\Omega} = 7.96; U_s = \frac{U_r}{Q} = \frac{400 \text{ kV}}{7.96} = 50.27 \text{ kV}$$

1.3) The reactor should be adjusted to **4.053 mH** and the output voltage of the source transformer needs to be **50.27 kV**

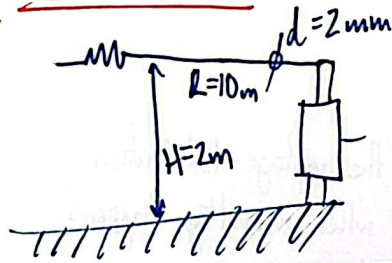
1.4) Disadvantages of cascaded transformers

- The lower stage is heavily loaded
- Low efficiency
- Insulation problems may occur since the voltage distribution among different stages is nonuniform when overvoltage happens

Advantages of ~~resonant~~ the series resonant circuit

- Improved voltage waveform by the elimination of unwanted resonances and attenuation of harmonics in the power supply
- The required power to run the test is lower than the actual power in the main test circuit
- Heavy power arcs cannot develop if a failure of the test occurs, only the load capacitance will be discharged

Question 2



$$Z_L = \ln \left[\frac{2l}{d} \sqrt{\frac{\sqrt{1+(2H/l)^2} - 1}{\sqrt{1+(2H/l)^2} + 1}} \right] \cdot \frac{1}{2\pi} \sqrt{\frac{\mu_0}{\epsilon_0}}$$

~ 60

$$Z_L \sim \ln \left(\frac{2 \cdot 10}{0.002} \sqrt{\frac{\sqrt{1+(2 \cdot 2/10)^2} - 1}{\sqrt{1+(2 \cdot 2/10)^2} + 1}} \right) \cdot 60 = 453 \Omega$$

2.1) R_d for a critically damped response needs to be 453 Ω

$$\text{Ref. coeff} = \frac{Z_L - Z_S}{Z_L + Z_S} = \frac{453 - 300}{453 + 300} = 0.203$$

2.2) Since the reflection coefficient is 0.203 the resulting waveform will show an underdamped response

2.3) To measure fast transients such as lightning voltages, high ohmic resistive dividers become impractical due to stray capacitances. Low ohmic resistive dividers aren't suitable either due to the high voltage requirement. Only capacitive or damped capacitive dividers should be used.

Question 4

$$C_N = 37.12 \text{ pF}$$

$$R_3 = 65 \Omega$$

$$R_4 = 1000/\pi$$

$$C_4 = 300 \text{ nF}$$

$$f = 50 \text{ Hz}$$

$$\delta = \arctan(\omega R_4 C_4) = \left(2\pi 50 \cdot \frac{1000}{\pi} \Omega \cdot 300 \text{ nF} \right)$$

$$= \arctan(0.03) = 0.02999 \text{ rad} = 1.718^\circ$$

$$C = C_N \cdot \frac{R_4}{R_3} = 37.12 \text{ pF} \cdot \frac{1000/\pi}{65} = 181.78 \text{ pF}$$

4.1) The dielectric loss angle is 1.718° and the capacitance is measured to be 181.78 pF

$$\tan \delta = \omega R_S C_S \Rightarrow R_S = \frac{\tan \delta}{\omega C_S} = \frac{0.004}{2\pi 50 \cdot 120 \text{ pF}} = 106.103 \text{ k}\Omega$$

$$I = U \left(j\omega C_S + \frac{1}{R_S} \right) = 10 \text{ kV} \cdot \left(j\omega 120 \text{ pF} + \frac{1}{106.103 \text{ k}\Omega} \right) =$$

$$I_{\text{NAD}} = 94.25 \text{ mA} \angle 0.229^\circ$$

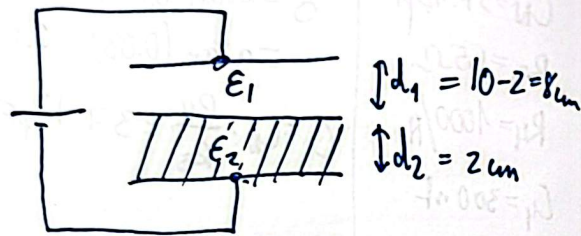
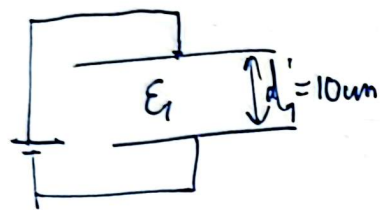
$$P_{\text{loss}} = I^2 R = (94.25 \text{ mA})^2 \cdot 106.103 \text{ k}\Omega = 942.5 \text{ W}$$

$$P_{\text{loss}} = \omega C V^2 = 2\pi 50 \cdot 120 \text{ pF} \cdot (10 \text{ kV})^2 = 3.77 \text{ W}$$

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4.2) The current would be 94.25 mA $\angle 0.229^\circ$ and the loss would be of 3.77 W

4.3) The measuring resistor R_3 and variable capacitor C_4 can be connected into the HV potential

Question 5

$$E_0 = \frac{V}{d_1} = \frac{200 \text{ kV}}{10 \text{ cm}} = 2 \text{ MV/m}$$

$$\epsilon_1 = 1$$

$$\epsilon_2 = 2.25$$

5.1) The electric field in the air between the electrodes is 2 MV/m

$$V = E_1 \cdot d_1 + E_2 \cdot d_2$$

$$E_1 = \frac{V}{d_2 + d_1 \left(\frac{\epsilon_1}{\epsilon_2} \right)} = \frac{200 \text{ kV}}{2 \text{ cm} + 8 \text{ cm} \left(\frac{1}{2.25} \right)} = 2.25 \text{ MV/m}$$

$$E_2 = \frac{V}{d_2 + d_1 \left(\frac{\epsilon_2}{\epsilon_1} \right)} = \frac{200 \text{ kV}}{2 \text{ cm} + 8 \text{ cm} \left(\frac{2.25}{1} \right)} = 1 \text{ MV/m}$$

5.2) The electric field in the remaining air is 2.25 MV/m and the electric field in the material is 1 MV/m

5.3) E_1 and E_2 would remain the same values since their thickness hasn't changed.

5.4) With more ~~ins~~ thickness of the insulation material, E_1 (air) would increase and E_2 (material) would decrease

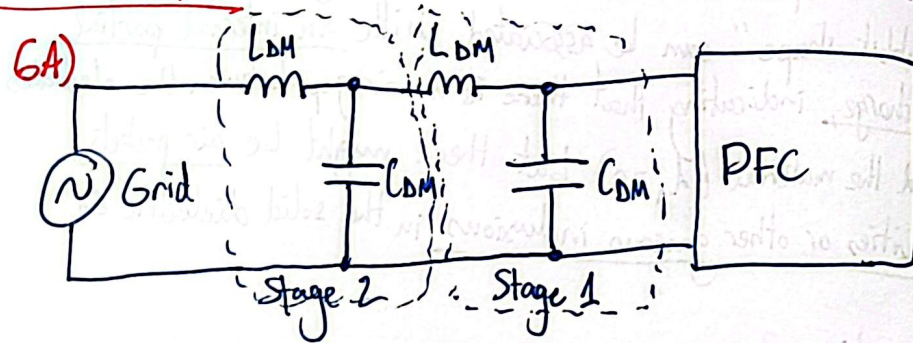
5.5) That pattern, close to the zero crossings and with a "rabbit shape" can be associated with an internal partial discharge, indicating that there is no air gap between the electrodes and the material ($\epsilon_1 = \phi$) but there might be air pockets, cavities or other gaseous inclusions in the solid dielectric ϵ_2

Question 3

- According to Paschen's law, at high pressures there will be many collisions, but since the gas will be more dense (smaller λ) ~~the mean free path~~ and v will be lower, hence increasing the breakdown voltage.

- Vacuum means pressure $\sim \phi$, so no atoms would make $V_b \rightarrow \infty$.

Question 6



- The PFC acts as a current source, which is considered high impedance. Hence, it must be matched with a low impedance (the capacitor)
- The grid acts as a voltage source (low impedance). It must be matched with a high impedance (the inductor)

6B) $AH = \text{Measurement} - \text{Limit} + \text{Margin} \rightarrow 6 \text{ dB}$

→ there are multiple peaks. Usually by bringing the first peak down, the rest will comply

→ $AH = 131.7 - 66 + 6 = 71.7 \text{ dB}$

$$AH(f) = |(1 + j2\pi f L_{DM} C_{DM})^2 + (j2\pi f)^2 L_{DM} C_{DM}|$$

Since it's a high frequency ($> 150 \text{ kHz}$) it can be simplified,

$$AH(f) = (2\pi f)^4 \cdot L_{DM}^2 C_{DM}^2 \Rightarrow L_{DM} = \sqrt{\frac{AH}{(2\pi f)^4 C_{DM}^2}}$$

SLA

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8 $\Rightarrow L_{DM} = \sqrt{\frac{10^{\frac{71.7}{20}}}{(2\pi \cdot 160 \text{ kHz})^4 \cdot (330 \text{ nF})^2}} = 185.94 \mu\text{H}$ 2025 1068

The closest compliant E-6 and E-12 values would be $220 \mu\text{H}$ or a E-24 value of $200 \mu\text{H}$ could be chosen.

6C) CISPR 11-Class A is used in establishments other than domestic areas, such as medical devices

CISPR 11-Class B is used in domestic establishment, such as kitchen appliances

6D)

→ Again, we try to bring down the first peak

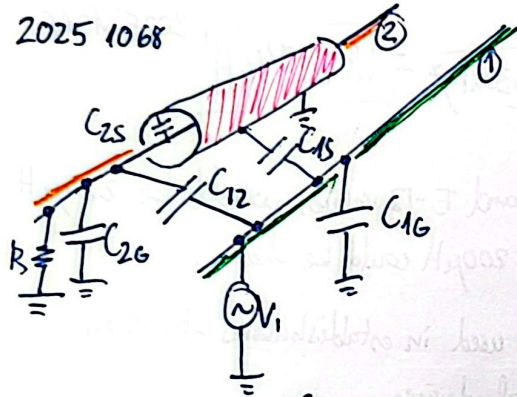
$$AH = 138.6 - 110 + 6 = 34.6 \text{ dB}$$

→ We no longer can use the simplification, because $f < 150 \text{ kHz}$

$$AH(f) = |(1 + j2\pi f)^2 L_{DM} C_{DM} + 1|^2 + (j2\pi f)^2 L_{DM} C_{DM}$$

→ Solved with CAS calculator, $f = 40 \text{ kHz}$, $AH = 10^{\frac{34.6}{20}}$, $L_{DM} = 200 \mu\text{H}$

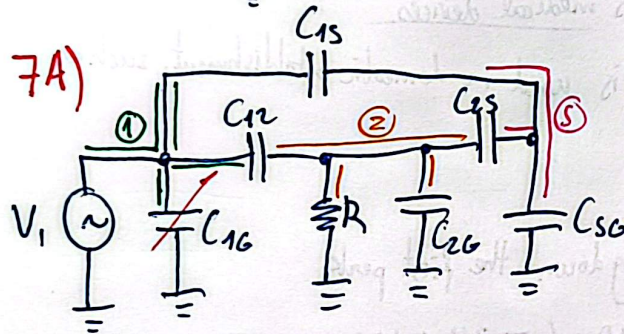
$C_{DM} = 705.53 \text{ nF} \rightarrow 820 \text{ nF for E-12, 750 nF for E-24}$



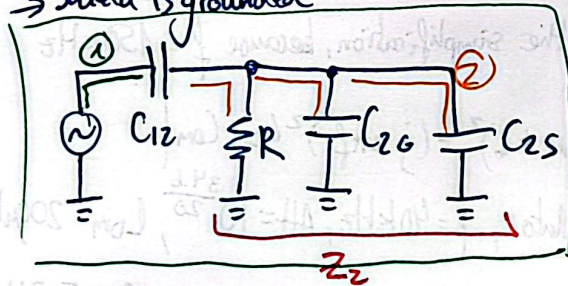
Question 7

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7A)



→ Shield is grounded



$$V_2 = V_1 \cdot \frac{Z_2}{Z_2 + Z_{C12}} \quad Z_{C12} = \frac{1}{j\omega C_{12}} \quad Z_2 = \frac{1}{\frac{1}{j\omega C_{26}} + \frac{1}{j\omega C_{25}} + \frac{1}{R}}$$

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$$V_2 = V_1 \cdot \frac{\frac{1}{\frac{1}{j\omega C_{26}} + \frac{1}{j\omega C_{25}} + \frac{1}{R}}}{\frac{1}{\frac{1}{j\omega C_{26}} + \frac{1}{j\omega C_{25}} + \frac{1}{R}} + \frac{1}{j\omega C_{12}}}$$

→ Simplified with CAS calculator

$$V_2 = V_1 \cdot \frac{j\omega R C_{12}}{j\omega R (C_{12} + C_{26} + C_{25}) + 1} = V_1 \cdot \frac{j\omega \cdot \frac{C_{12}}{C_{12} + C_{25} + C_{26}}}{j\omega + \frac{1}{R \cdot (C_{12} + C_{26} + C_{25})}}$$

$$\rightarrow \text{if } R \ll \frac{1}{j\omega (C_{12} + C_{26} + C_{25})} \rightarrow V_2 = V_1 \cdot j\omega R C_{12}$$

$$\rightarrow \text{if } R \gg \frac{1}{j\omega (C_{12} + C_{26} + C_{25})} \rightarrow V_2 = V_1 \cdot \frac{C_{12}}{C_{12} + C_{26} + C_{25}}$$

7B)

7B.1)

First we calculate $\frac{1}{j\omega (C_{12} + C_{26} + C_{25})} = \frac{1}{2\pi \cdot 100\text{kHz} \cdot (5\text{pF} + 15\text{pF} + 12\text{pF})}$

$$= 11.37\text{k}\Omega \gg 20\Omega = R \Rightarrow V_2 = V_1 \cdot j\omega R \cdot C_{12} = 10\text{V} \cdot 2\pi \cdot 100\text{kHz} \cdot 20\Omega \cdot 5\text{pF} = 628\mu\text{V}$$

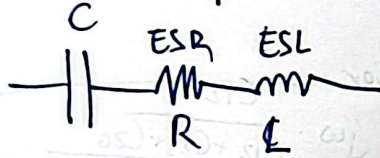
$$V_2 = 10\text{V} \cdot 2\pi \cdot 100\text{kHz} \cdot 20\Omega \cdot 5\text{pF} = 628\mu\text{V}$$

7B.2) $\frac{1}{j\omega (C_{12} + C_{26} + C_{25})} \Big|_{1\text{MHz}} = 1.137\text{k}\Omega \gg 20\Omega = R \Rightarrow \text{Same eq.}$

$$V_2 = 2\text{V} \cdot 2\pi \cdot 1\text{MHz} \cdot 20\Omega \cdot 5\text{pF} = 1.26\text{mV}$$

Question 8

8A) It is a capacitor.



8B)

The resistance is obtained directly at the resonant frequency

$$\boxed{R = 46 \text{ m}\Omega}$$

$$C = \frac{1}{2\pi \cdot f \cdot Z_c} = \frac{1}{2\pi \cdot 10 \text{ kHz} \cdot 65.23 \Omega} = \boxed{244 \text{ nF}}$$

$$L = \frac{Z}{2\pi \cdot f} = \frac{2.45 \Omega}{2\pi \cdot 10 \text{ MHz}} = \boxed{39 \text{ nH}}$$